

Infrastructure Investment and Jobs Act funding allocation

Population explains most differences in total funding, but per-resident allocations vary sharply

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DATA 608: Knowledge and Visual Analytics
CUNY School of Professional Studies
Funding data as of March 2023

\$196.0B

funding in the course dataset

Data and analytical scope

02

\$196.0B

Total course-provided funding

56

Geographic jurisdictions

51

Election comparison jurisdictions

Equity benchmark

A jurisdiction receives its population share of the geographic funding pool. The allocation ratio equals funding share divided by population share.

Population

July 2022 Census estimates cover the states, District of Columbia, and Puerto Rico. Official 2020 Census counts cover four Island Areas.

Political comparison

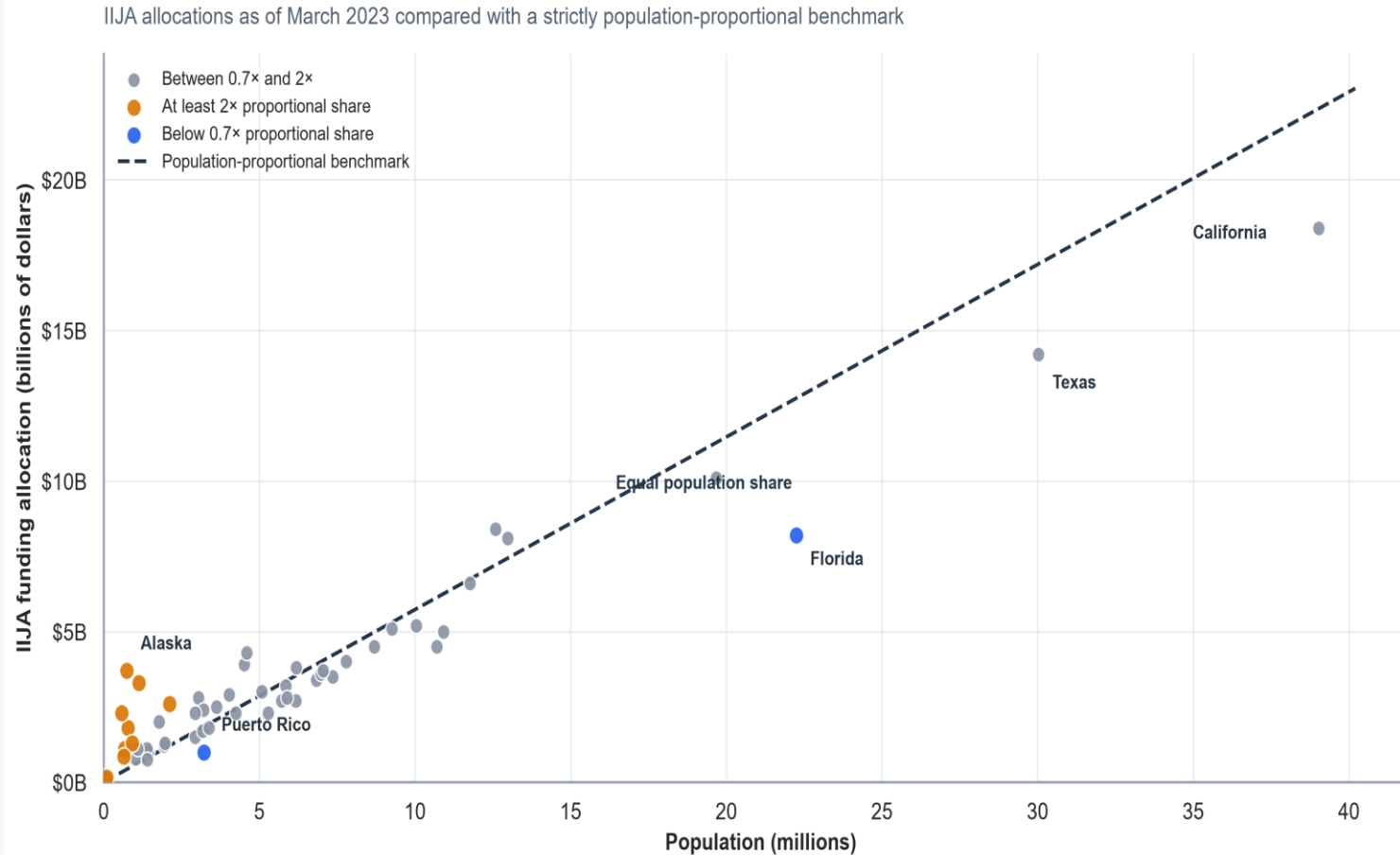
Official FEC statewide 2020 results identify 26 Biden-won and 25 Trump-won jurisdictions. Territories do not enter this comparison.

Scope decision

The \$3.0 billion Tribal Communities category remains separate because its population overlaps state and territorial populations.

Population strongly tracks total funding

Funding rises with population, but several small jurisdictions receive outsized shares



Sources: Course-provided IIJA allocation data (March 2023); U.S. Census Bureau 2022 estimates and 2020 Island Areas counts.
Note: Tribal Communities are excluded because their population overlaps state and territorial populations.

Equity scope: 50 states, District of Columbia, and five territories; Tribal Communities excluded

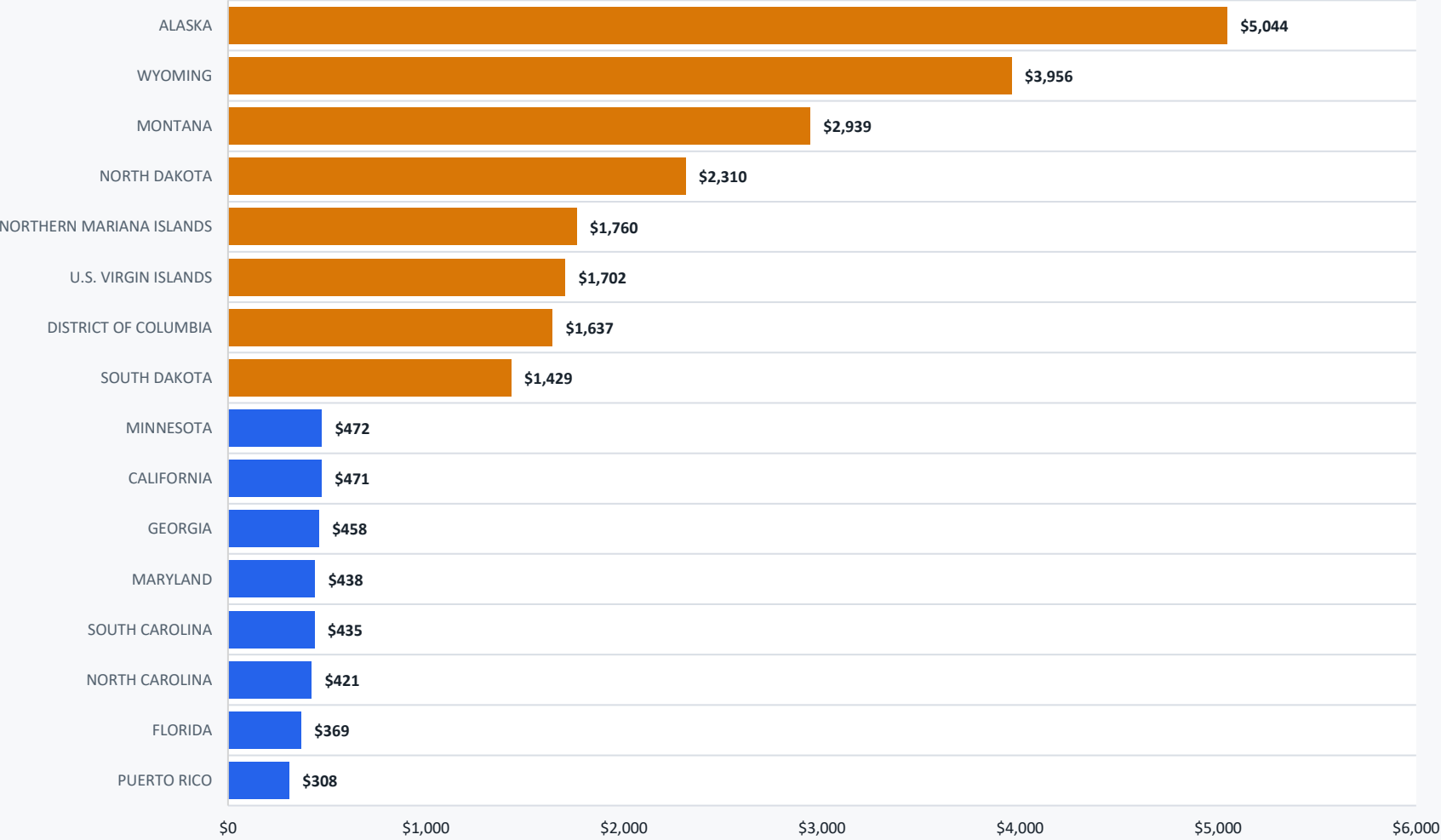
Pearson correlation: 0.966

Large jurisdictions receive more total funding. Several small jurisdictions still receive much more than their population-proportional share.

Interpretation

Population explains most of the total-dollar pattern. It does not explain the full distribution of dollars per resident.

Per-resident funding ranges from \$308 to \$5,044



16.4x

Highest-to-lowest difference

The wide range confirms that the allocation is not strictly proportional to population.

Orange: eight highest values. Blue: eight lowest values. Common horizontal scale.

The allocation does not favor Biden-won jurisdictions



13.4% higher

in Trump-won jurisdictions on a population-weighted basis

Mann-Whitney test $p = 0.172$

Population-adjusted model $p = 0.341$

Neither comparison reaches conventional statistical significance.

The evidence supports two conclusions

1 Allocation is not strictly population proportional

Population strongly predicts total funding, yet per-resident allocations differ by more than sixteenfold across the observed jurisdictions.

2 The data shows no preference for Biden-won jurisdictions

Trump-won jurisdictions received more per resident. Neither the distributional test nor the population-adjusted model found a significant political difference.

What this analysis cannot determine

- 1 Funding allocations do not equal completed spending.
- 2 Population does not capture infrastructure need or statutory funding minimums.
- 3 A statewide election winner is a simplified measure of political alignment.
- 4 The observational data cannot establish intent or causation.

Sources and AI-use disclosure

Funding

Course-provided IJJA FUNDING AS OF MARCH 2023.xlsx

Population

U.S. Census Bureau, NST-EST2022-POP

U.S. Census Bureau, 2020 Island Areas population counts

Election

Federal Election Commission, Federal Elections 2020

AI-use disclosure

ChatGPT was used to assist with data-cleaning review, analysis, visualization refinement, and editorial review. I reviewed and validated all code, calculations, visualizations, interpretations, and final conclusions.

Supporting notebook, processed data, and reproducible analysis are available through the GitHub repository linked in the submission comments.